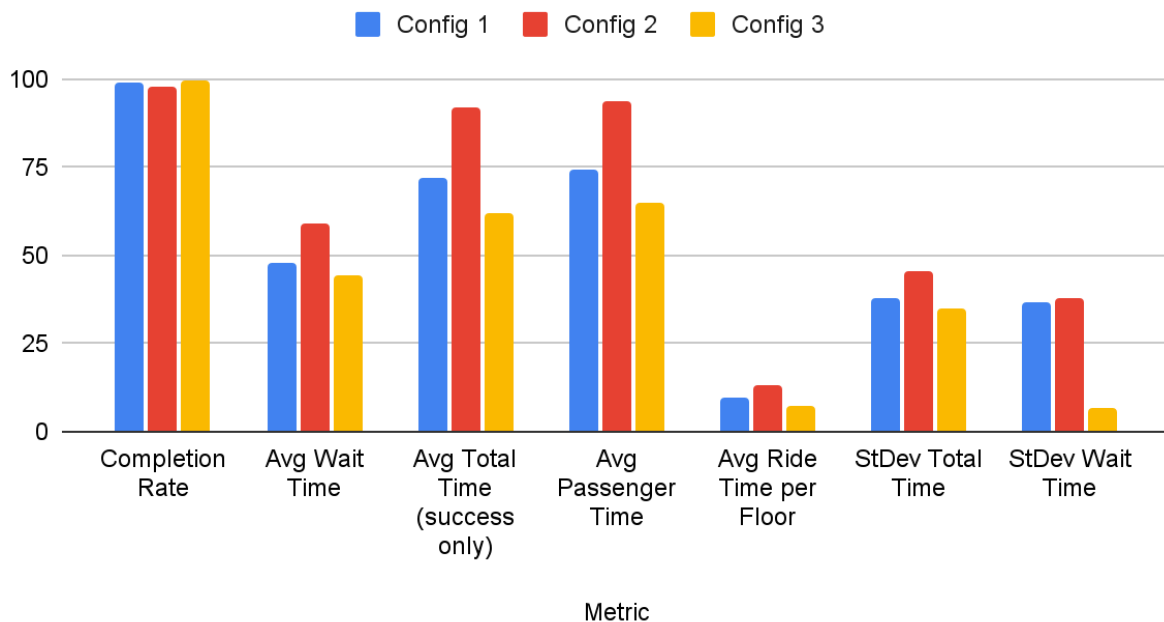


Elevator Analysis

During analysis, we were asked to compare three different elevators with three different configurations. These different configurations were an elevator with (Capacity=15, floorTicks=5, doorTicks=2, and passPerTick=3), an elevator with (Capacity=20, floorTicks=7, doorTicks=3, and passPerTick=4), and an elevator with (Capacity=10, floorTicks=3, doorTicks=2, and passPerTick=3). Using our code to run tests on these three, we were able to get our data on the time it takes to wait and the total time for each passenger of each elevator. The data can be found here: [FullElevatorTestCfg1PassData](#) We chose to use a variety of metrics to analyze the performance of the elevators, including completion rate, average wait time, average total time, average time by passenger, average ride time per floor, the standard deviation of the total time, and the standard deviation for the wait time.

Config 1, Config 2 and Config 3



Note that we only have 1 chart that has all the data for our analysis across the 5 metrics.

We can see that all three elevators have exceptional completion rates, with rates of 99.2%, 98.0%, and 99.6%, respectively. Although only a small percentage of passengers give up, it should be noted that, when compared to Config 3, passengers are twice as likely to give up if the elevator is Config 1 and five times as likely to give up when it is Config 2. In terms of average wait time, we can see that Config 2 has the longest wait time of 59 ticks, whereas Config 1 has 48 ticks and Config 3 has an average of just 44 ticks. The pattern of Config 3 have a generally quicker wait time, and Config 2 having a generally longer one stays consistent when we analyze the total time (Config 3, 1, 2 in order of speed), the weighted passenger average time (to account for larger groups), and the average ride time per floor. These last 4 statistics (Wait Time, Total Time, Passenger Time, and Floor Time) are quite similar, but are important to analyze nonetheless in case one elevator was, for example, being very efficient with larger

groups but not smaller ones, or a certain elevator had a short wait time but long ride time. Finally, when we look at the standard deviation for the total time, we can see that Elevator config 3 is slightly more consistent with the total time a passenger will spend on their journey when compared to Elevator configs 1 and 2, but is tremendously consistent when it comes to the average wait time experienced by a passenger. By analyzing all the data, and seeing that Config 3 is the quickest, most reliable, and most consistent elevator, we feel confident in **recommending Config 3 to be used as an Elevator configuration.** Despite Config 3 having the smallest capacity, it was not an issue in terms of passenger efficiency, as it had quick doorTicks and passPerTick. While it is true that a large group could not take advantage of an Elevator in configuration 3, groups of such size are few and far between; **the elevator should be configured to serve the majority of people, which config 3 does best.**